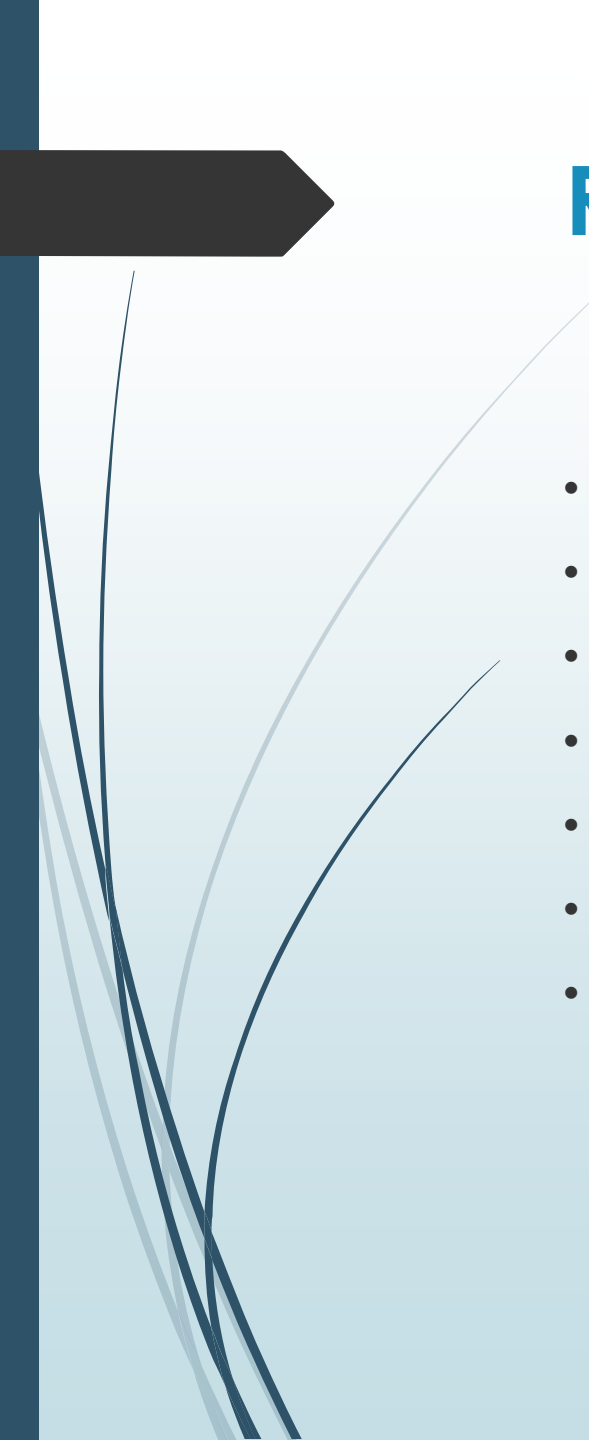




Raritone Internship

Role: AI/ML Intern

Name:Akkireddy Hasini



Research Completed

- Real-time pose estimation using MediaPipe & OpenPose
- Human body segmentation techniques
- 3D body mesh generation using SMPL
- 2D garment warping and try-on pipelines
- Cloth simulation and fabric physics
- AI-based clothing recommendation systems
- Dataset collection for fashion AI



Models Explored

AI Models Used

Model

Purpose

Mediapipe

33 body landmarks in real time

OpenPose

135 keypoints, multi-person

SMPL

Full 3D body mesh generation

SCHP

Body part segmentation

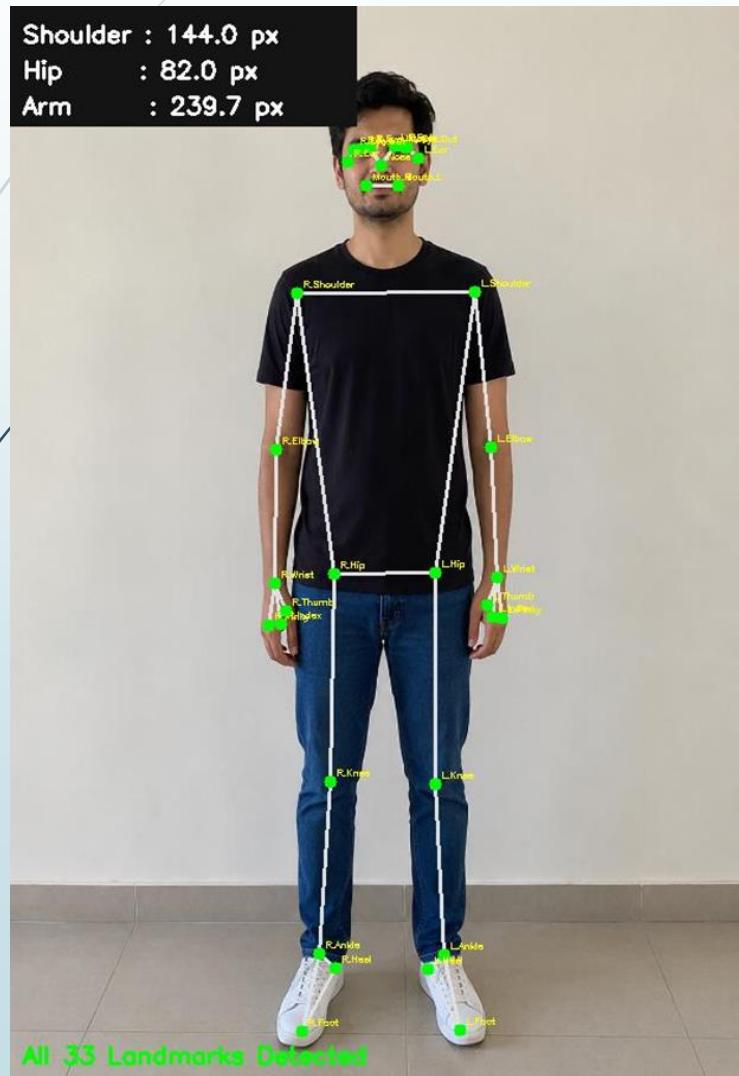
HR-VITON

2D garment try-on

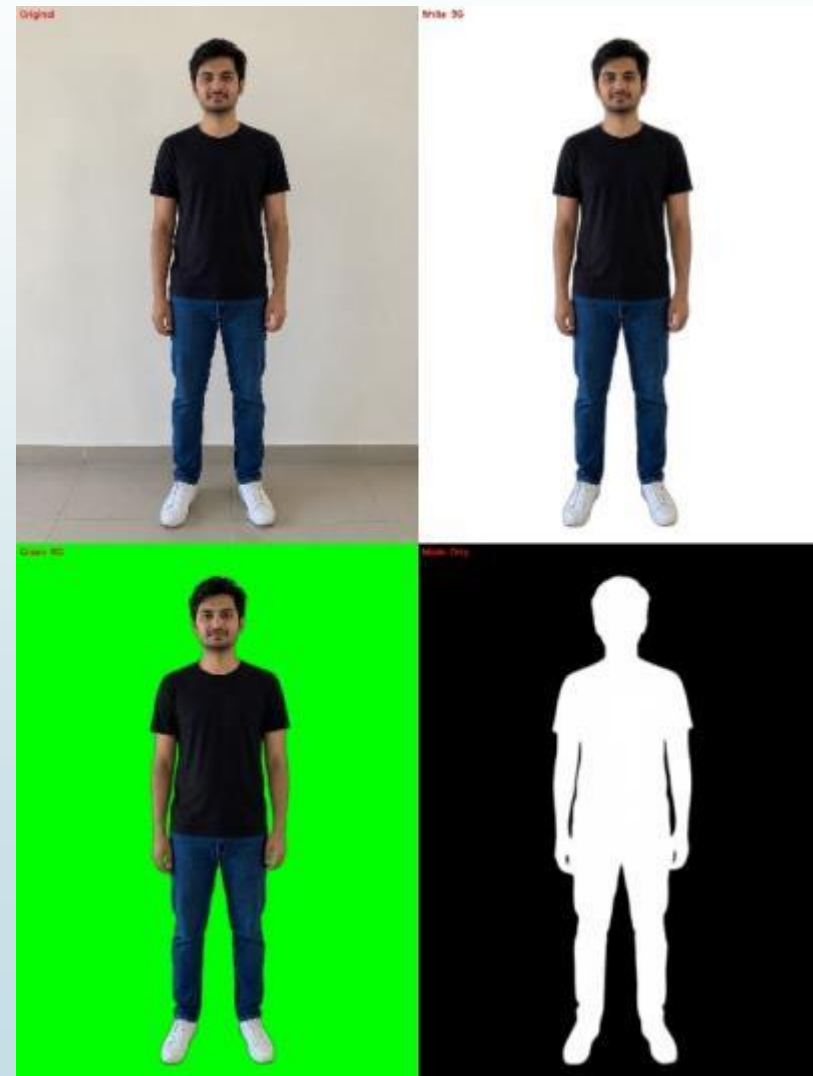
Detectron2

High quality person masking

Pose Detection



Segmentation





Datasets Collected

Datasets Researched & Organized

Dataset

VITON-HD

DeepFashion

CLOTH3D

MPV

Custom

Contents

13K person + garment pairs

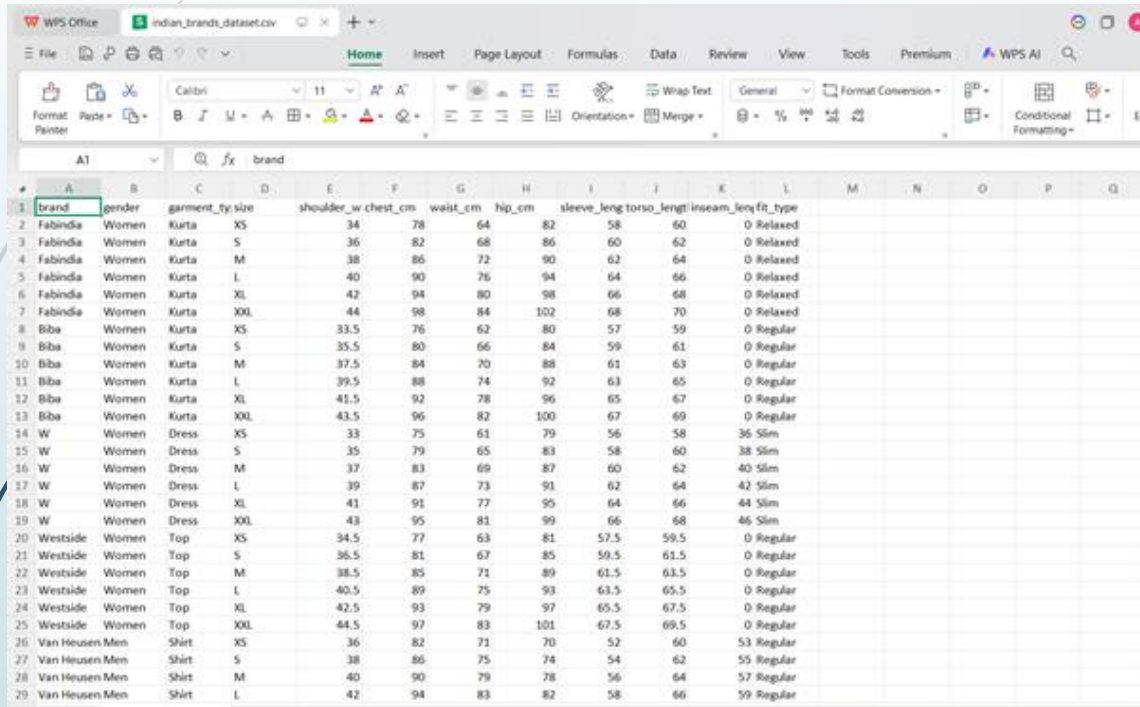
800K+ clothing images

3D garment meshes

Multi-pose try-on pairs

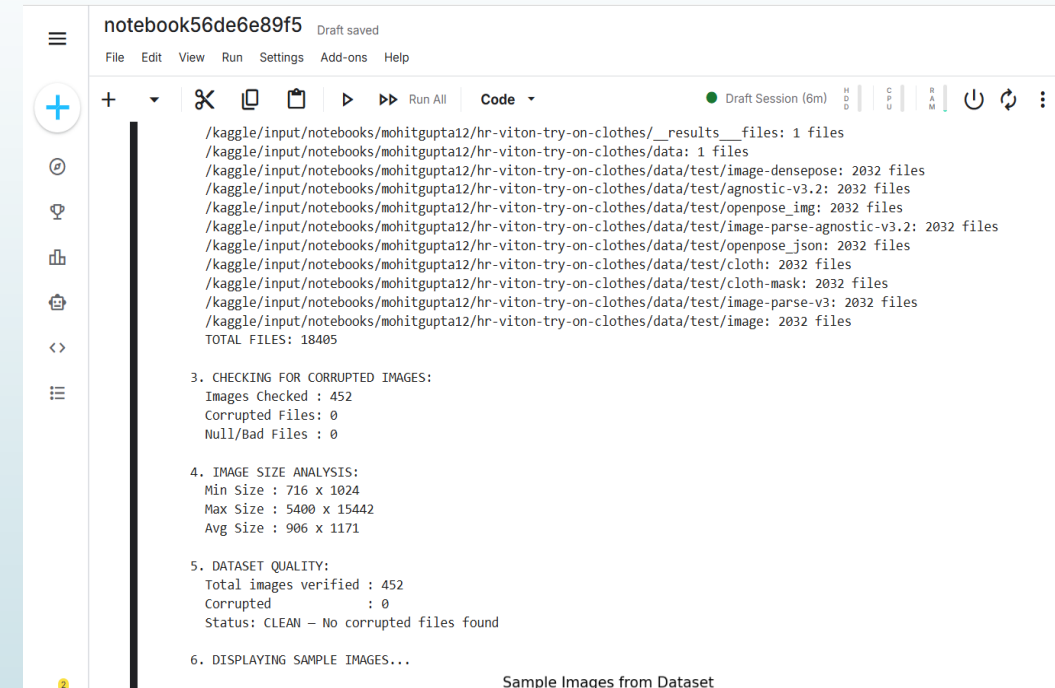
Indian ethnic wear, diverse
body types

Custom



#	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	brand	gender	garment_ty	size	shoulder_w	chest_cm	waist_cm	hip_cm	sleeve_long	torso_len	inseam	len	fit_type				
2	Fabinda	Women	Kurta	XS	34	78	64	82	58	60	0	Relaxed					
3	Fabinda	Women	Kurta	S	36	82	68	86	60	62	0	Relaxed					
4	Fabinda	Women	Kurta	M	38	86	72	90	62	64	0	Relaxed					
5	Fabinda	Women	Kurta	L	40	90	76	94	64	66	0	Relaxed					
6	Fabinda	Women	Kurta	XL	42	94	80	98	66	68	0	Relaxed					
7	Fabinda	Women	Kurta	XXL	44	98	84	102	68	70	0	Relaxed					
8	Biba	Women	Kurta	XS	33.5	76	62	80	57	59	0	Regular					
9	Biba	Women	Kurta	S	35.5	80	66	84	59	61	0	Regular					
10	Biba	Women	Kurta	M	37.5	84	70	88	61	63	0	Regular					
11	Biba	Women	Kurta	L	39.5	88	74	92	63	65	0	Regular					
12	Biba	Women	Kurta	XL	41.5	92	78	96	65	67	0	Regular					
13	Biba	Women	Kurta	XXL	43.5	96	82	100	67	69	0	Regular					
14	W	Women	Dress	XS	33	75	61	79	56	58	36	Slim					
15	W	Women	Dress	S	35	79	65	83	58	60	38	Slim					
16	W	Women	Dress	M	37	83	69	87	60	62	40	Slim					
17	W	Women	Dress	L	39	87	73	91	62	64	42	Slim					
18	W	Women	Dress	XL	41	91	77	95	64	66	44	Slim					
19	W	Women	Dress	XXL	43	95	81	99	66	68	46	Slim					
20	Westside	Women	Top	XS	34.5	77	63	81	57.5	59.5	0	Regular					
21	Westside	Women	Top	S	36.5	81	67	85	59.5	61.5	0	Regular					
22	Westside	Women	Top	M	38.5	85	71	89	61.5	63.5	0	Regular					
23	Westside	Women	Top	L	40.5	89	75	93	63.5	65.5	0	Regular					
24	Westside	Women	Top	XL	42.5	93	79	97	65.5	67.5	0	Regular					
25	Westside	Women	Top	XXL	44.5	97	83	101	67.5	69.5	0	Regular					
26	Van Heusen	Men	Shirt	XS	36	82	71	70	52	60	53	Regular					
27	Van Heusen	Men	Shirt	S	38	86	75	74	54	62	55	Regular					
28	Van Heusen	Men	Shirt	M	40	90	79	78	56	64	57	Regular					
29	Van Heusen	Men	Shirt	L	42	94	83	82	58	66	59	Regular					

VITON-HD



```
notebook56de6e89f5 Draft saved
File Edit View Run Settings Add-ons Help

+ - ✂ 📄 📌 ▶ ⏮ Run All Code ▾ Draft Session (6m) H D C P U R A M 🔌 🔄 ⋮

/kaggle/input/notebooks/mohitgupta12/hr-viton-try-on-clothes/_results__files: 1 files
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/kaggle/input/notebooks/mohitgupta12/hr-viton-try-on-clothes/data/test/image-parse-v3: 2032 files
/kaggle/input/notebooks/mohitgupta12/hr-viton-try-on-clothes/data/test/image: 2032 files
TOTAL FILES: 18405

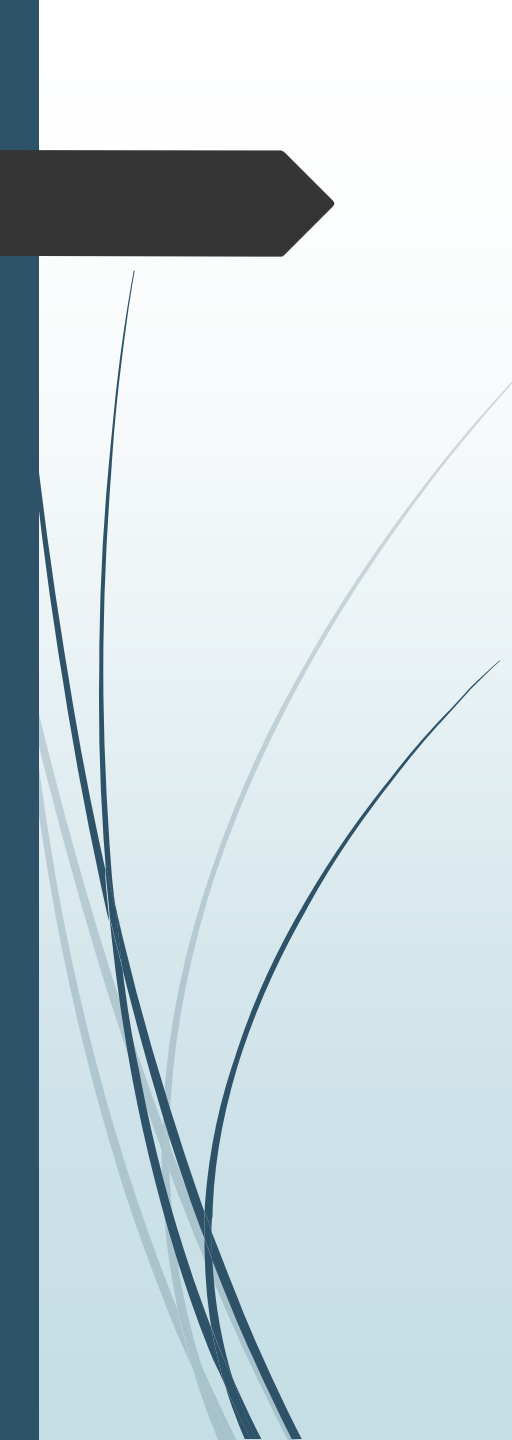
3. CHECKING FOR CORRUPTED IMAGES:
Images Checked : 452
Corrupted Files: 0
Null/Bad Files : 0

4. IMAGE SIZE ANALYSIS:
Min Size : 716 x 1024
Max Size : 5400 x 15442
Avg Size : 906 x 1171

5. DATASET QUALITY:
Total images verified : 452
Corrupted : 0
Status: CLEAN - No corrupted files found

6. DISPLAYING SAMPLE IMAGES...
```

Sample Images from Dataset



2D Virtual Try-On Findings

2D Try-On Pipeline

► User Photo + Clothing → Pose Estimation → Human Parsing → Clothing Mask → TPS Warping → HR-VITON → Output

What Worked:

Clean results on front-facing standard poses

Realistic fabric folds and texture with HR-VITON

Fast near real-time processing on GPU

What Needs Work:

Unusual poses cause misalignment

Baggy clothing warps poorly

Edge artifacts at clothing boundaries

Original



Background Removed

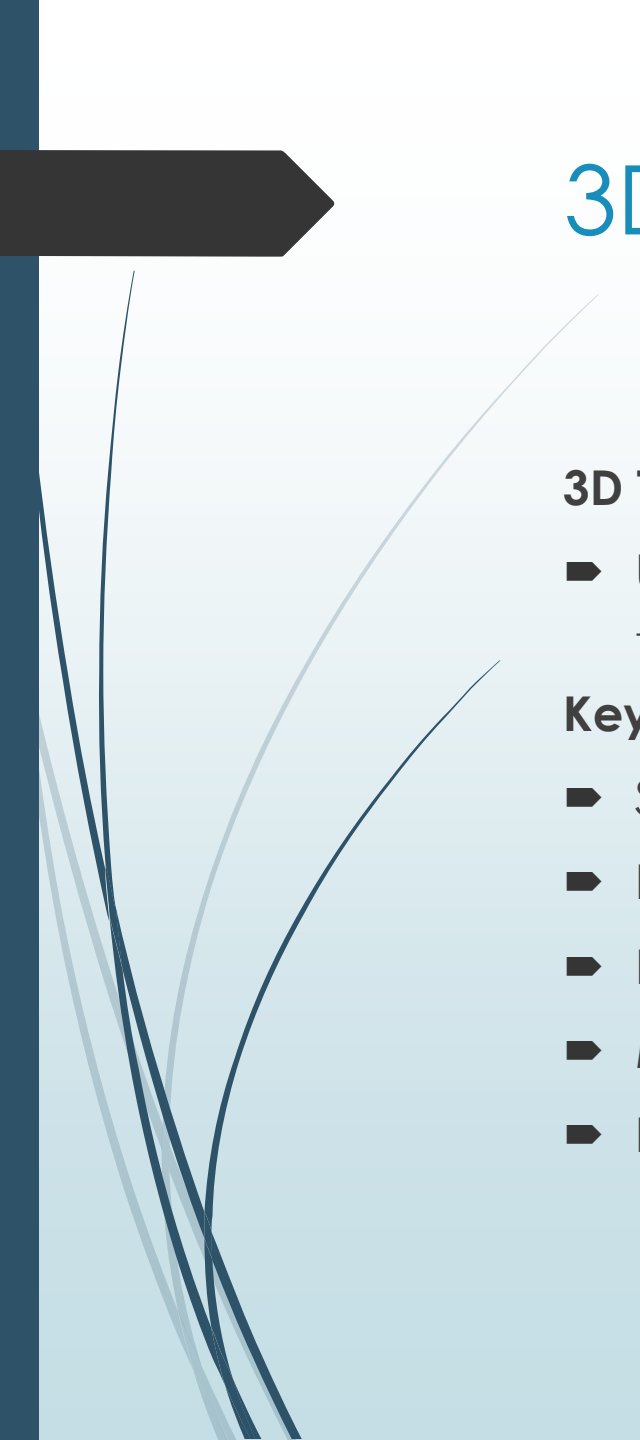


Pose Detection



Try-On Output





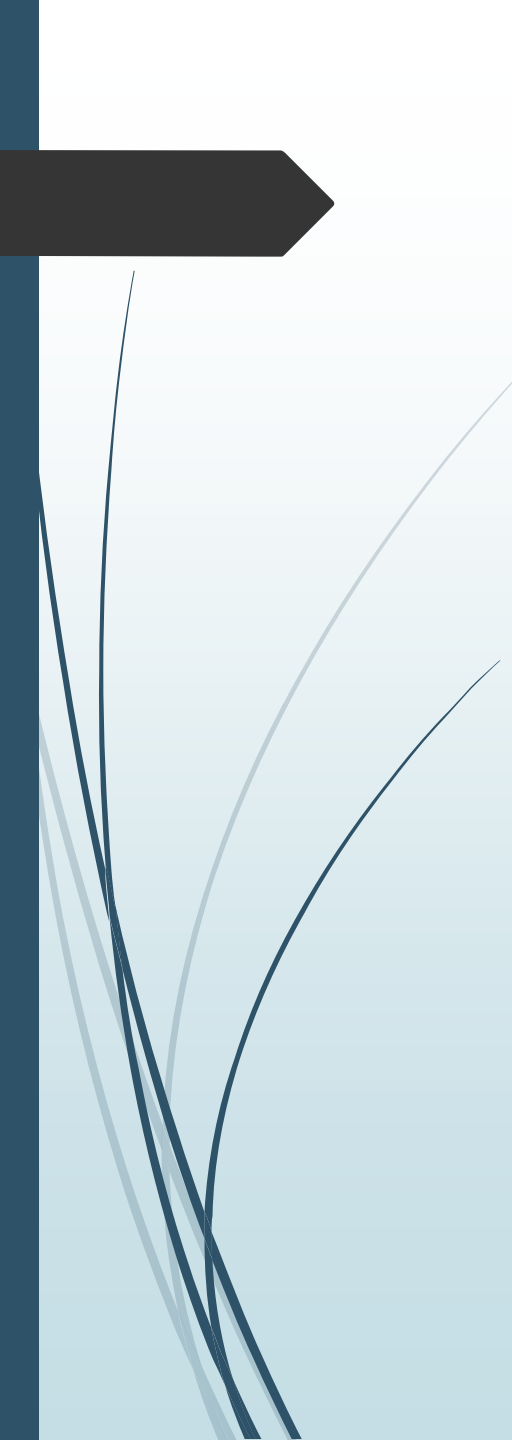
3D Virtual Try-On Findings

3D Try-On Research

- User Measurements → SMPL Body Mesh → Avatar Fitting → Cloth Simulation → Real-Time Rendering → 3D Try-On

Key Findings:

- SMPL accurately estimates body shape from basic measurements
- PBD cloth simulation gives realistic fabric drape
- Real-time 30 FPS achievable with optimized mesh
- MediaPipe skeleton can drive live avatar movement
- Plus-size body types need more diverse training data

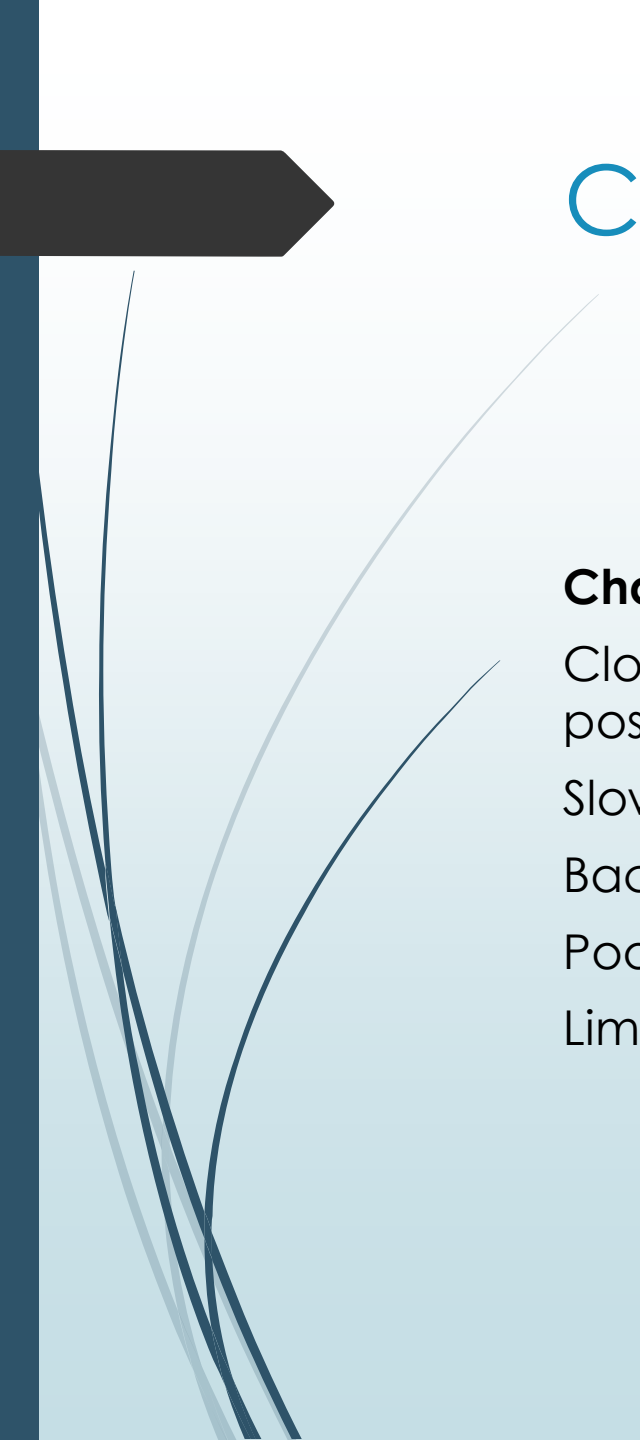


AI Recommendation System Ideas

User Profile → Body Type Classification → Size Prediction → Style Matching → Fit Scoring → Ranked Suggestions → One-Click Try-On

Key Ideas:

- Predict correct size per brand from body measurements
- Learn style preferences from try-on and purchase history
- Rate fit quality of each garment (score 1–10)
- Suggest alternatives when size is unavailable
- Get smarter with every interaction



Challenges & Solutions

Challenge

Clothing misalignment on unusual poses

Slow processing speed

Background artifacts

Poor results on baggy clothing

Limited ethnic wear in datasets

Solution

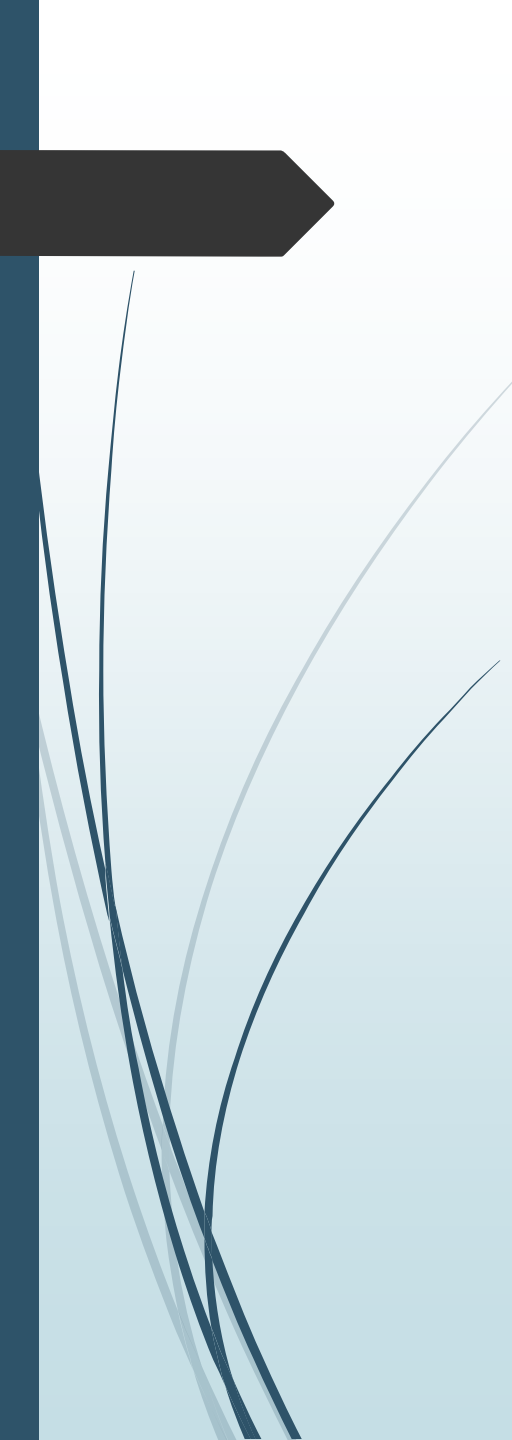
More diverse pose training data

MobileNet + input resizing

Detectron2 for cleaner masks

Improved TPS control points

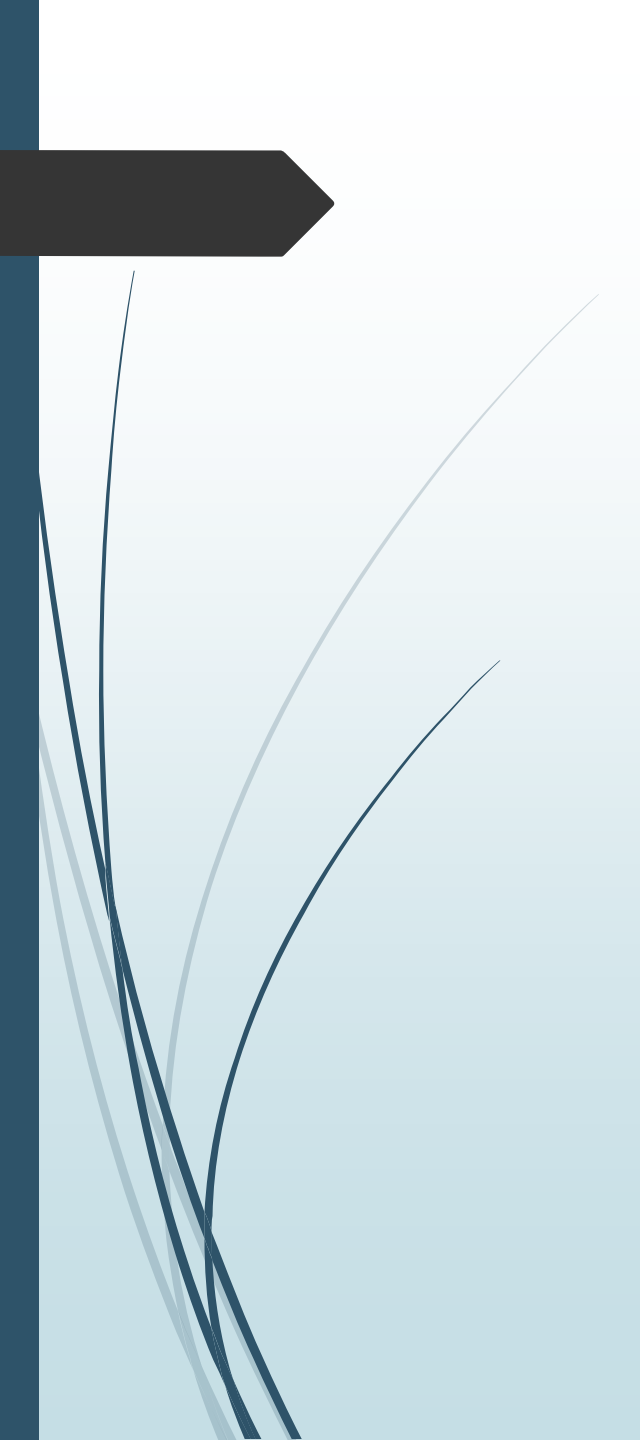
Custom local dataset collected



Future Scope

Live AR try-on on mobile camera

- Body measurements from 2 photos automatically
- Full 3D avatar with real cloth physics
- Multi-garment try-on — full outfit at once
- Smarter size recommendation per brand
- Deploy as web or mobile app
- Smart mirror for physical retail stores



Thank You